22.2.2 Exercises - Photoelectric effect 2

1.

a. Describe the photoelectric effect.

.....

.....[2]

b. Data for the work function energy Φ of two metals are shown in Fig.

	Φ / J
sodium	3.8×10^{-19}
zinc	5.8×10^{-19}

Light of wavelength 420nm is incident on the surface of each of the metals.

i. State what is meant by a photon.

.....

.....[2]

ii. Calculate the energy of a photon of the incident light.

energy = ... J [2]

iii. State whether photoelectric emission will occur from each of the metals.

sodium.....

zinc: .....[2]

2.

a. One wavelength of electromagnetic radiation emitted from a mercury vapour lamp is 436 nm. Calculate the photon energy corresponding to this wavelength.

energy = ... J [2]

b. Light from the lamp in (a) is incident, separately, on the surfaces of caesium and tungsten metal. Data for the work function energies of caesium and tungsten metal are given in Fig.

metal	work function energy / eV
caesium	1.4
tungsten	4.5

Calculate the threshold wavelength for photoelectric emission from

i. caesium,

threshold wavelength = ... nm [2]

ii. tungsten.

threshold wavelength = ... nm [1]

c. Use your answers in (b) to state and explain whether the radiation from the mercury lamp of wavelength 436nm will give rise to photoelectric emission from each of the metals.

caesium: .....

tungsten: .....[2]

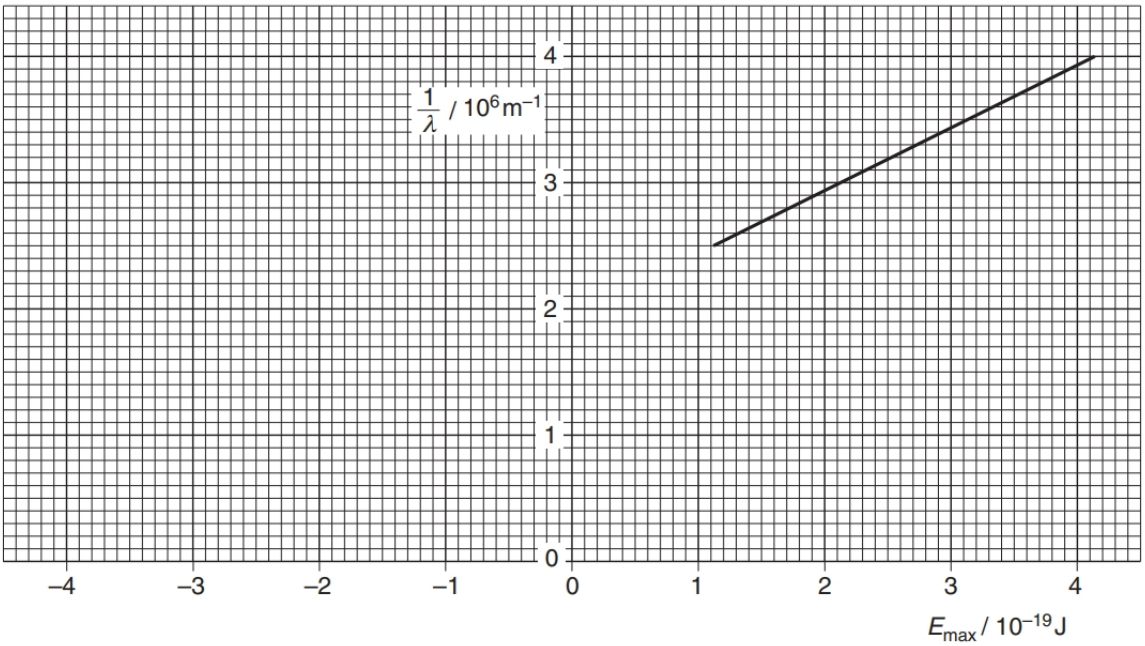
3. An explanation of the photoelectric effect includes the terms photon energy and work function energy.

a. State what is meant by the work function energy of a metal.

.....

.....[2]

b. In an experiment to investigate the photoelectric effect, a student measures the wavelength λ of the light incident on a metal surface and the maximum kinetic energy E_{max} of the emitted electrons. The variation with E_{max} of 1/λ is shown in Fig.



i. The work function energy of the metal surface is Φ.

State an equation, in terms of λ, Φ and Emax, to represent conservation of energy for the photoelectric effect. Explain any other symbols you use.

.....

.....[2]

ii. Use your answer in (i) and Fig. to determine

a) the work function energy Φ of the metal surface,

Φ = ... J [2]

b) a value for the Planck constant.

Planck constant = ... J s [3]

4.

a. Briefly describe two phenomena associated with the photoelectric effect that cannot be explained using a wave theory of light.

1. .....

2. .....[2]

b. The maximum energy E_{MAX} of electrons emitted from a metal surface when illuminated by light of wavelength λ is given by the expression

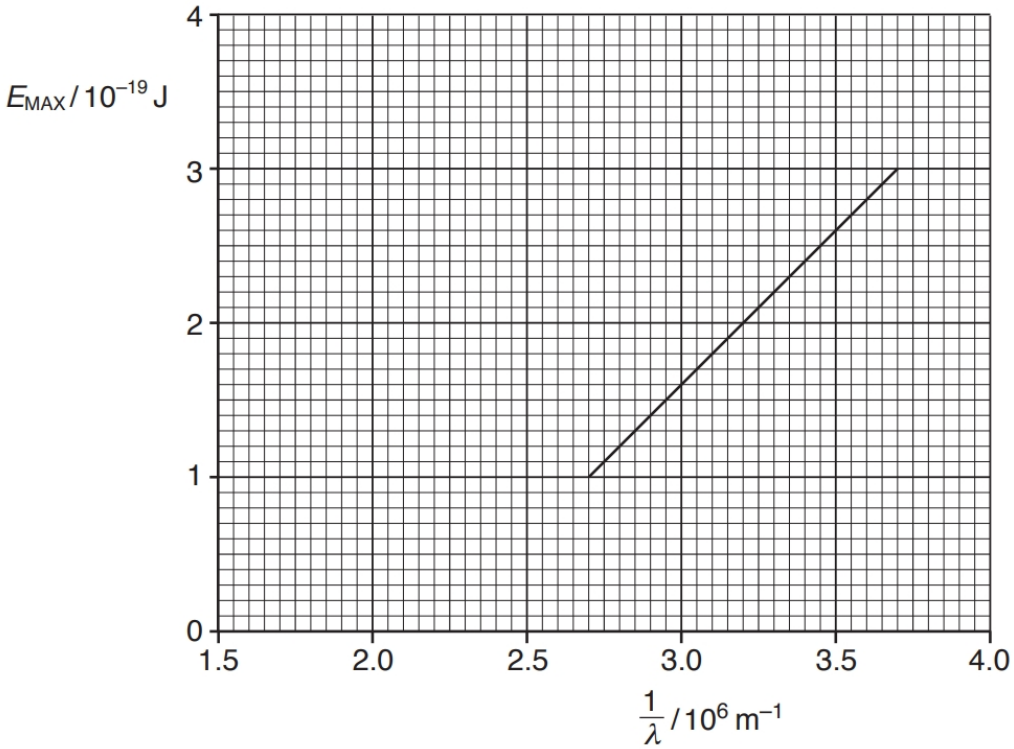
$$E_{MAX} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

where h is the Planck constant and c is the speed of light.

i. Identify the symbol λ_0.

.....[1]

ii. The variation with $\frac{1}{\lambda}$ of E_{MAX} for the metal surface is shown in Fig.



a) Use Fig. to determine the magnitude of λ0.

λ_0 = ... m [1]

b) Use the gradient of Fig. to determine a value for the Planck constant h.

h = ... Js [3]

c. The metal surface in (b) becomes oxidised.

Photoelectric emission is still observed but the work function energy is increased.

On Fig., draw a line to show the variation with $\frac{1}{\lambda}$ of E_{MAX} for the oxidised surface. [2]